

## SAFETY DATA SHEET

Creation Date 24-Nov-2010

Revision Date 25-Mar-2024

Revision Number 3

### 1. Identification

**Product Name** Hexafluorosilicic acid, 35% w/w aqueous solution

**Cat No. :** L14829

**Synonyms** Fluorosilicic acid; Hexafluorosilicic acid

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Skin Corrosion/Irritation

Serious Eye Damage/Eye Irritation

Category 1 B

Category 1

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Causes severe skin burns and eye damage

**Precautionary Statements****Prevention**

Do not breathe dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

Immediately call a POISON CENTER or doctor/physician

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing

**Skin**

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower  
Wash contaminated clothing before reuse

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

**Ingestion**

IF SWALLOWED: Rinse mouth. DO NOT induce vomiting

**Storage**

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component          | CAS No     | Weight % |
|--------------------|------------|----------|
| Water              | 7732-18-5  | 65-75    |
| Fluorosilicic acid | 16961-83-4 | 25-35    |

### 4. First-aid measures

|                                            |                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                         | Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.                                                                                                                                                               |
| <b>Skin Contact</b>                        | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Immediate medical attention is required.                                                                                                                                                  |
| <b>Inhalation</b>                          | Remove to fresh air. Get medical attention. If not breathing, give artificial respiration.                                                                                                                                                                                                      |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician immediately.                                                                                                                                                                                                                                           |
| <b>Most important symptoms and effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                                                                                                                                                                                                           |

### 5. Fire-fighting measures

**Suitable Extinguishing Media** Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

**Unsuitable Extinguishing Media** No information available

**Flash Point** No information available  
**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

### Hazardous Combustion Products

Silicon dioxide. Gaseous hydrogen fluoride (HF).

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
3

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Avoid contact with skin and eyes. Use personal protective equipment as required. Ensure adequate ventilation.

**Environmental Precautions** See Section 12 for additional Ecological Information. Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Take up mechanically and collect in suitable container for disposal. After cleaning, flush away traces with water.

## 7. Handling and storage

**Handling** Ensure adequate ventilation. Wear personal protective equipment/face protection. Avoid contact with skin and eyes. Do not breathe mist/vapors/spray.

**Storage.** Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Corrosives area.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component          | ACGIH TLV                  | OSHA PEL                             | NIOSH                                                     | Mexico OEL (TWA)           |
|--------------------|----------------------------|--------------------------------------|-----------------------------------------------------------|----------------------------|
| Fluorosilicic acid | TWA: 2.5 mg/m <sup>3</sup> | (Vacated) TWA: 2.5 mg/m <sup>3</sup> | IDLH: 250 mg/m <sup>3</sup><br>TWA: 2.5 mg/m <sup>3</sup> | TWA: 2.5 mg/m <sup>3</sup> |

### Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures** Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations

and safety showers are close to the workstation location.

### **Personal Protective Equipment**

|                                 |                                                                                                                                                                                                                                                   |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye/face Protection</b>      | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.                                                                       |
| <b>Skin and body protection</b> | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                                                                                         |
| <b>Respiratory Protection</b>   | Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. |
| <b>Recommended Filter type:</b> | Particulates filter conforming to EN 143.                                                                                                                                                                                                         |
| <b>Hygiene Measures</b>         | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                            |

## **9. Physical and chemical properties**

|                                               |                            |
|-----------------------------------------------|----------------------------|
| <b>Physical State</b>                         | Liquid                     |
| <b>Appearance</b>                             | Clear                      |
| <b>Odor</b>                                   | pungent                    |
| <b>Odor Threshold</b>                         | No information available   |
| <b>pH</b>                                     | 1-1.2 @ 20°C 10 g/L aq.sol |
| <b>Melting Point/Range</b>                    | No data available          |
| <b>Boiling Point/Range</b>                    | No information available   |
| <b>Flash Point</b>                            | No information available   |
| <b>Evaporation Rate</b>                       | No information available   |
| <b>Flammability (solid,gas)</b>               | Not applicable             |
| <b>Flammability or explosive limits</b>       |                            |
| <b>Upper</b>                                  | No data available          |
| <b>Lower</b>                                  | No data available          |
| <b>Vapor Pressure</b>                         | No information available   |
| <b>Vapor Density</b>                          | No information available   |
| <b>Specific Gravity</b>                       | 1.270                      |
| <b>Solubility</b>                             | No information available   |
| <b>Partition coefficient; n-octanol/water</b> | No data available          |
| <b>Autoignition Temperature</b>               | No information available   |
| <b>Decomposition Temperature</b>              | No information available   |
| <b>Viscosity</b>                              | No information available   |
| <b>Molecular Formula</b>                      | H2 F6 Si                   |
| <b>Molecular Weight</b>                       | 144.09                     |

## **10. Stability and reactivity**

|                                         |                                                 |
|-----------------------------------------|-------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available      |
| <b>Stability</b>                        | Stable under normal conditions.                 |
| <b>Conditions to Avoid</b>              | Incompatible products.                          |
| <b>Incompatible Materials</b>           | Strong oxidizing agents                         |
| <b>Hazardous Decomposition Products</b> | Silicon dioxide, Gaseous hydrogen fluoride (HF) |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.        |
| <b>Hazardous Reactions</b>              | None under normal processing.                   |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

##### Oral LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

##### Dermal LD50

Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

##### Vapor LC50

Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.

#### Component Information

| Component          | LD50 Oral                          | LD50 Dermal | LC50 Inhalation              |
|--------------------|------------------------------------|-------------|------------------------------|
| Water              | -                                  | -           | -                            |
| Fluorosilicic acid | 125 mg/kg (Rat)<br>430 mg/kg (Rat) | Not listed  | LC50 = 1.11 mg/L ( Rat ) 1 h |

#### Toxicologically Synergistic Products

No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

#### Irritation

Causes burns by all exposure routes

#### Sensitization

No information available

#### Carcinogenicity

The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component          | CAS No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------|------------|------------|------------|------------|------------|------------|
| Water              | 7732-18-5  | Not listed | Not listed | Not listed | Not listed | Not listed |
| Fluorosilicic acid | 16961-83-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

#### Mutagenic Effects

No information available

#### Reproductive Effects

No information available.

#### Developmental Effects

No information available.

#### Teratogenicity

No information available.

#### STOT - single exposure

None known

#### STOT - repeated exposure

None known

#### Aspiration hazard

No information available

#### Symptoms / effects, both acute and delayed

Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

#### Endocrine Disruptor Information

No information available

#### Other Adverse Effects

The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains. Do not flush into surface water or sanitary sewer system. Large amounts will affect pH and harm aquatic organisms.

| Component          | Freshwater Algae | Freshwater Fish                                                                                          | Microtox   | Water Flea |
|--------------------|------------------|----------------------------------------------------------------------------------------------------------|------------|------------|
| Fluorosilicic acid | Not listed       | LC50: = 28.7 mg/L, 96h static (Pimephales promelas)<br>LC50: = 65 mg/L, 96h static (Poecilia reticulata) | Not listed | Not listed |

#### Persistence and Degradability

No information available

**Bioaccumulation/ Accumulation** No information available.

**Mobility** No information available.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN1778  
 Proper Shipping Name FLUOROSILICIC ACID  
 Hazard Class 8  
 Packing Group II

#### TDG

UN-No UN1778  
 Proper Shipping Name FLUOROSILICIC ACID  
 Hazard Class 8  
 Packing Group II

#### IATA

UN-No UN1778  
 Proper Shipping Name FLUOROSILICIC ACID  
 Hazard Class 8  
 Packing Group II

#### IMDG/IMO

UN-No UN1778  
 Proper Shipping Name FLUOROSILICIC ACID  
 Hazard Class 8  
 Packing Group II

### 15. Regulatory information

#### United States of America Inventory

| Component          | CAS No     | TSCA | TSCA Inventory notification -<br>Active-Inactive | TSCA - EPA Regulatory<br>Flags |
|--------------------|------------|------|--------------------------------------------------|--------------------------------|
| Water              | 7732-18-5  | X    | ACTIVE                                           | -                              |
| Fluorosilicic acid | 16961-83-4 | X    | ACTIVE                                           | -                              |

#### **Legend:**

**TSCA** US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

'-' - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

#### International Inventories

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component          | CAS No     | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|--------------------|------------|-----|------|-----------|-------|------|------|------|-------|----------|
| Water              | 7732-18-5  | X   | -    | 231-791-2 | X     | X    |      | X    | X     | KE-35400 |
| Fluorosilicic acid | 16961-83-4 | X   | -    | 241-034-8 | X     | X    | X    | X    | X     | KE-18550 |

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product does not contain any chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)** Not applicable

**Clean Air Act** Not applicable

**OSHA - Occupational Safety and Health Administration** Not applicable

**CERCLA**

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355). There may be specific reporting requirements at the local, regional, or state level pertaining to releases of this material.

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component          | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|--------------------|---------------|------------|--------------|----------|--------------|
| Water              | -             | -          | X            | -        | -            |
| Fluorosilicic acid | X             | X          | -            | -        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): N  
DOT Marine Pollutant N  
DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

**Other International Regulations**

**Mexico - Grade** No information available

**Authorisation/Restrictions according to EU REACH**

| Component          | CAS No     | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------|------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Water              | 7732-18-5  | -                                                                   | -                                                                             | -                                                                                                     |
| Fluorosilicic acid | 16961-83-4 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component          | CAS No     | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------|------------|----------|------------------------------|---------------------------|--------------------------------------------|
| Water              | 7732-18-5  | Listed   | Not applicable               | Not applicable            | Not applicable                             |
| Fluorosilicic acid | 16961-83-4 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

## Contains component(s) that meet a 'definition' of per &amp; poly fluoroalkyl substance (PFAS)?

Not applicable

## Other International Regulations

| Component          | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Water              | 7732-18-5  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Fluorosilicic acid | 16961-83-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

## Prepared By

Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

## Creation Date

24-Nov-2010

## Revision Date

25-Mar-2024

## Print Date

25-Mar-2024

## Revision Summary

New emergency telephone response service provider.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**